Environmental variability and bet-hedging

Learning objectives

1. Connect environmental variability to changes in lambda
2. Compare populations with a range of growth rates and sensitivities to enviornmental variability
3. Model a population with a seed bank

last lecture: unrestricted population growth

modeled environmental stocahsticity by making lambda vary from year to year

this is a workaround for measuring and modeling actual environmental variables

Spatial environmental variability

on small scales, patchiness makes populations grow or shrink

- example: light & understory species
- λ varies in space
- accumulation in good patches
- lattice models, cells affect population growth

draw 4x4 lattice, random patches and plants

lambda=0.9 or 1.1 to indicate where populations growing and shrinking

Temporal environmental variability

- example: climate
- $\lambda > 1$ in wet years
- $\lambda < 1$ in dry years

species can avoid bad patches by dispersing into good patches - how might a species avoid bad years?

Bet-hedging and seed banks

temporal analog to dealing with bad years

- evolve to cope with environmental variability *in other words, variability is a selective agent*
- bet-hedging: lower variation in fitness, but lower overall fitness *growth*
- mechanism: seed banks
- low germination rate: more seeds in seed bank, spread risk *some seeds in reserve in case its a bad year to germinate*
- high germination rate: fewer seeds in seed bank, risky

similar to gambling, if plants make small bets, keeping most chips (seeds) in reserve, less risky; if you make large bets, more risky, and it might pay out (like in a good year) or you might lose everything (go locally extinct)

Demonstration of bet-hedging

bet-hedging is a good strategy in the long run

- low variance, low mean more beneficial than high variance, high mean

Year	Population 1	Population 2
1	0.2	0.8
2	3.0	1.2
3	0.4	0.9
4	2.5	1.1
arithmetic mean	0.42	0
geometric mean	-0.12	-0.01

Modeling seed banks

in lab, we'll model density independent growth with a seed bank

- λ = per capita growth rate
- N = number of seeds *for annual plant model*
- g = germination rate *of seeds produced by adults*
- s = survival of seeds in seed bank

$$N_{t+1} = N(1-g)s + \lambda gN$$

- includes the seeds that go into the seed bank
- first term = seeds in seed bank
- second term = new seeds